Pagemark 0.13.1 Android mobile audit

Version 0.13.1

Commit e42cc26

2026-10-14

Revision 1.2

Mobile Application Release Audit - Pagemark (Android)

Version 0.13.1 (2022), commit e42cc26 = tag v0.13.1. Platform: Android only (iOS out of scope, scaffolding). Environment: production candidate. Artifacts: Pagemark-0.13.1-arm64.apk sha256 3a0f7e78...351be9; armv7.apk f5151fdc...ded0b4; play.aab fd5ee9b5...016b0734. Evidence mode: artifact + static + runtime smoke on one Android 17 x86_64 emulator (arm64 APK, ABI-translated) + unit/widget suite. No real phone. Identity gate: PASS. Audited 2026-10-14. Store policy checked 2026-10-14: developer.android.com/google/play/requirements/target-sdk (page dated 2026-10-01). MASVS: control groups used by name; IDs not cited.

DECISION

- GitHub sideload APKs: CONDITIONAL GO. No blocker, no S3+ finding.
- Google Play upload: CONDITIONAL NO-GO until the Data safety form is checked (MOB-PRV-0002) and a real-device pass is done.

Coverage: ~24 controls tested (PASS/PASS-WARN), 0 FAIL; NOT TESTED or MANUAL REVIEW: fresh install/removal, navigation/state, TalkBack, device matrix, clipboard/recents, real-phone performance, Data safety. See coverage.md. Scores (coverage): Security 99 (~70%), Privacy 98 (~50%), Store 97 (~40%), Reliability 100 (~30%), UX/Perf not scored (emulator only). Counts: blockers 0, S4 0, S3 0, S2 1, S1 4, S0 1 (accepted).

Findings

- MOB-PRV-0002 S2 (store) Data safety draft omits ML Kit scanner telemetry - manual review
- MOB-PRV-0001 S1 Privacy policy permission list and "Drive backups" wording stale
- MOB-BLD-0003 S1 APK v2-only signature, no rotation lineage
- MOB-AND-0004 S1 sqlite3_flutter_libs +eol; no vuln scan
- MOB-STORE-0005 S1 PLAY_STORE.md checklist stale, feature graphic missing
- MOB-SEC-0006 S0 locked DB unencrypted - owner-accepted

Passed highlights

Not debuggable; backup and device-transfer off; no cleartext or custom TLS; https hosts only; Play AAB has no install permission; target 36 and 16 KB alignment OK; same signing cert on both APKs; PIN PBKDF2 150k + salt + lockout; backup AES-256-GCM; analyze clean; 295 tests pass. Runtime: 0 crashes/ANRs, upgrade keeps data, offline launch and 2.0 text scale render cleanly, 0 OSV vulnerabilities in 187 packages.

Missing for a full GO

Device pass (MANUAL_TEST_PLAN.md: timer, folder backup to cloud, fingerprint, notifications, TalkBack, updater, upgrade from 0.13.0), logcat/clipboard/recents check, scanner traffic capture, real-phone performance numbers, and verification that the 0.13.1 fixes for PGM-300/302/320 work on a device.

Details: findings/*.md, coverage.md, artifact/.

MOB-AND-0004 - Dependency sqlite3_flutter_libs is on its "+eol" release; OSV scan found no known CVEs

Severity S1 | **Status** open | **Confidence** medium | **Basis** static | **Blockers** []

The package version string marks end of life; the bundled libsqlite3.so will not receive upstream updates through it. OSV batch query of all 187 locked pub packages returned no vulnerabilities (artifact/sbom-pub.txt); the concern is maintenance only. Remediation: move to the maintained sqlite3 native-assets path (sqlite3 ^3.x is already a dependency) and rerun the scan. Verify: pubspec has no +eol package and the OSV scan is clean.

MOB-BLD-0003 - Release APK signed with APK Signature Scheme v2 only (no v3/rotation lineage)

Severity S1 | **Status** open | **Confidence** high | **Basis** artifact | **Blockers** []

apksigner: v1 false, v2 true, v3 false. Valid for minSdk 24. Without v3 a compromised or lost key cannot be rotated for sideload users, and CLAUDE.md already warns that losing the key blocks in-place upgrades. Play App Signing covers the Play build, not the GitHub APKs. Remediation: keep the keystore backed up (NAS script exists); optionally sign with v3 so a rotation lineage can be added later. Verify: apksigner verify shows v3 true.

MOB-PRV-0001 - Privacy policy permission list and Drive wording do not match the shipped manifest

Severity S1 | **Status** open | **Confidence** high | **Basis** artifact | **Blockers** []

Merged manifest also holds VIBRATE, WAKE_LOCK, RECEIVE_BOOT_COMPLETED, FOREGROUND_SERVICE, USE_FINGERPRINT (from plugins/WorkManager/notifications); the policy lists none of them. The Locked-library paragraph still says exports include "Drive backups", but Drive backup was removed in 0.8.2. Remediation: list the extra permissions with one-line reasons (or strip unused ones with tools:node="remove" if confirmed unused, e.g. FOREGROUND_SERVICE), replace "Drive backups" with "folder backups". Verify: aapt2 dump badging vs the policy list.

MOB-PRV-0002 - Play Data safety draft does not address ML Kit barcode scanner telemetry

Severity S2 | **Status** open | **Confidence** medium | **Basis** static | **Blockers** [store]

PRIVACY.md admits Google's ML Kit "may report anonymous technical information to Google", yet the Data safety answers say nothing is collected except lookups. Google can flag a form that omits SDK collection. Unverified at runtime (no traffic capture), so MANUAL REVIEW. Remediation: capture the scanner's network traffic once on a device, then declare diagnostics/device identifiers accordingly (or confirm none), and align both documents. Verify: traffic log + Play Console form.

MOB-SEC-0006 - Locked library is not encrypted at rest (owner decision)

Severity S0 | **Status** accepted-risk | **Confidence** high | **Basis** static | **Blockers** []

The PIN gates the UI; the SQLite file is plain inside app-private storage (backup, adb and cloud transfer disabled). Owner declined SQLCipher on 2026-10-12; recorded, not re-proposed. Exposure needs a rooted device or forensic access; the encrypted backup option covers exported copies.

MOB-STORE-0005 - PLAY_STORE.md open items are stale (refers to 0.7.1) and the feature graphic is still missing

Severity S1 | **Status** open | **Confidence** high | **Basis** static | **Blockers** []

Checklist still says verify 0.7.1 fixes and regenerate screenshots; 1024x500 graphic not made. Remediation: refresh the checklist for 0.13.x/1.0.0 and create the graphic. Verify: all listing assets present in Play Console.

Coverage - Pagemark 0.13.1+22, Android (sideload arm64 APK + play AAB), evidence mode: artifact + static + runtime smoke on Android 17 x86_64 emulator (arm64 APK, ABI-translated) + test suite

Method tags: [artifact] [static] [runtime-suite] (unit/widget tests only, no device). | Control | State | Evidence | |---|---|---| | 01 identity: pkg com.axialforge.pagemark, 0.13.1 (2022), target/compile 36, min 24, arm64 | PASS [artifact] | artifact/badging.txt | | 01 identity gate: tag v0.13.1 == HEAD e42cc26, tree clean, APK built after commit | PASS [static] | git rev-parse | | 02 not debuggable, no debuggable attr | PASS [artifact] | merged-manifest.txt | | 02 signing release key (CN=Pagemark), same cert on arm64+armv7, key.properties gitignored | PASS-WARN [artifact] MOB-BLD-0003 | signing.txt | | 02 R8 mapping present; no secrets/staging URLs found (hits are license/library strings) | PASS-WARN [artifact] | secret-smells.txt | | 02 16 KB page alignment of .so | PASS [artifact] | zipalign -P 16 OK | | 02 play AAB lacks REQUEST_INSTALL_PACKAGES | PASS [artifact] | AAB manifest grep | | 03 upgrade: 0.13.1 installed over existing emulator install, data kept (library shows Dune) | PASS [runtime] | rt-launch.png, rt-package.txt | | 03 fresh install, removal/reinstall, upgrade from a published release | NOT TESTED; migrations covered by suite [runtime-suite] | test.txt | | 05/08 text scale 2.0 on Library: no overflow or clipping | PASS [runtime] | rt-font2.png | | 11/20 launch in airplane mode, no crash; process kill then relaunch restores app | PASS [runtime] | rt-offline.png, logcat 0 fatal | | 07 back/rotation/state restore, deep paths | NOT TESTED | | | 04,06,08 (TalkBack),09 (device matrix: one emulator only) | NOT TESTED / MANUAL REVIEW (MANUAL_TEST_PLAN.md) | | | 10 performance: cold start 2.8-4.4 s, PSS 231 MB (emulator, ABI translation, not representative) | PASS-WARN [runtime] MANUAL REVIEW on real phone | rt-perf.txt | | 12 local storage: allowBackup=false, extraction rules exclude all | PASS [artifact] | merged-manifest.txt | | 12 locked DB unencrypted | accepted-risk MOB-SEC-0006 | | | 12/15 PIN PBKDF2-SHA256 150k, random 16B salt, attempt limiting, const-time compare; backup AES-256-GCM | PASS-WARN [static] | lib/lock.dart, crypto_backup.dart | | 12 logcat: 0 fatal/ANR, no PIN/title/token content from the app; run-as refused (not debuggable) | PASS [runtime] | rt-logcat.txt | | 12 clipboard, recents thumbnail (FLAG_SECURE on lock) | NOT TESTED (static only) | | | 13 network: only https hosts, no cleartext config, no custom TrustManager | PASS [static] | grep lib | | 14 backend | N/A (no backend, no accounts) | CLAUDE.md, lib grep | | 15 accounts / account deletion | N/A (no accounts) | | | 17 permissions (runtime-granted set matches manifest; Android 17 auto-grants ACCESS_LOCAL_NETWORK to this older target, not declared) : INTERNET, CAMERA, BIOMETRIC, POST_NOTIFICATIONS justified; extras | PASS-WARN MOB-PRV-0001 | badging.txt | | PLATFORM exported components: MainActivity (MAIN+SEND text, whitelisted), others permission-guarded or unexported | PASS-WARN [artifact] (untrusted-intent handling static only) | merged-manifest.txt | | 18 deps: 187 pub packages, OSV query: 0 known vulnerabilities; SBOM list; sqlite3_flutter_libs +eol | PASS-WARN [artifact] MOB-AND-0004 | sbom-pub.txt | | 19 tamper/resilience | N/A for threat model (local hobby app) | | | 21 telemetry: none declared; no analytics SDK | PASS [static] | pubspec.yaml | | 22 target API 36 meets Play rule (new apps/updates need 36 from 2026-08-31; page dated 2026-10-01) | PASS [artifact] | developer.android.com/google/play/requirements/target-sdk | | 22 Data safety, content rating, declarations | MANUAL REVIEW MOB-PRV-0002 | | | 23 listing claims vs behaviour | PASS-WARN [static] MOB-STORE-0005 | PLAY_STORE.md | | Suite: flutter analyze clean; flutter test 295 pass, 1 skip | PASS [runtime-suite] | analyze.txt, test.txt |

Appendix A - Scope, method and evidence index

Scope and environment

- App: Pagemark, `com.axialforge.pagemark`, version 0.13.1 (versionCode 2022), commit e42cc26 = tag v0.13.1, clean working tree.
- Platform: Android only. The iOS folder is scaffolding (needs a Mac) and is out of scope.
- Flavors: sideload (GitHub APK, self-updates, has REQUEST_INSTALL_PACKAGES) and play (AAB, no installer permission). Both audited from the build artifacts.
- Threat model: local-only personal app, no accounts, no backend. Data classes: reading log (user content), optional PIN-gated "locked library", backup files. Attackers considered: lost or stolen device, co-resident malicious app, network observer, malicious share intent. Reverse engineering and tamper resistance are out of profile (MASVS-RESILIENCE not required).
- Evidence modes used: artifact inspection, source review, unit and widget suite, and a runtime smoke on one Android 17 x86_64 emulator. The arm64 APK ran through ABI translation, so timings are not representative. No real phone was used.
- Tools: aapt2 and apksigner (build-tools 36.0.0), zipalign (16 KB check), adb, flutter analyze and flutter test, OSV batch query (api.osv.dev) for dependencies.
- OWASP MASVS control groups are named by group only. No MASWE or MASTG IDs are cited because they were not looked up at audit time.

Artifacts audited

File	Size (bytes)	SHA-256
Pagemark-0.13.1-arm64.apk	32,947,434	3a0f7e78b6081018119ed667206f453b4ed68f0dd8b41c3a5d93f6fcf0351be9
Pagemark-0.13.1-armv7.apk	29,161,302	f5151fdca3c9f003cc13f85d93cc89c46ddc9ec8f7c7f2e07f9eeece40ded0b4
Pagemark-0.13.1-play.aab	80,344,732	fd5ee9b589106beeae3599b85deb96e3a1b79c541b315900b6aca14e016b0734

Identity gate (PASS)

- Git: tag v0.13.1 resolves to HEAD e42cc2674f928e8f0063a910ef9332daff532825; `git status` clean; release files were written after the commit (19:02 vs 18:59).
- Manifest: package, versionName 0.13.1, versionCode 2022, compile and target SDK 36, minSdk 24, native code arm64-v8a (arm64 APK).
- Signing: one signer, CN=Pagemark, O=AxialForge, RSA 2048, certificate SHA-256 26b4a4234bc4346eb7e63ebee49c114cbdadd1fa10e60a5cc34f324d6b1bac44, identical on arm64 and armv7. Scheme: v2 only (see MOB-BLD-0003). The keystore and `key.properties` are not tracked by git.

Store policy sources checked

- Google Play target API requirement, https://developer.android.com/google/play/requirements/target-sdk, fetched 2026-10-14 (page last updated 2026-10-01): new apps and updates must target API 36 from 2026-08-31; Pagemark targets 36.
- Apple checks: not applicable (Android only).
- Data safety, content rating and account-deletion policies were not fetched; those controls are MANUAL REVIEW.

Evidence files (folder `artifact/`)

badging.txt, merged-manifest.txt, signing.txt, sha256.txt, file-list.txt, secret-smells.txt, manifest-flags.txt, analyze.txt, test.txt, sbom-pub.txt, rt-perf.txt, rt-logcat.txt, rt-launch.png, rt-offline.png, rt-font2.png.

Automated results

- `flutter analyze`: No issues found.
- `flutter test`: 295 passed, 1 skipped, 0 failed.
- Secret smell scan of the unpacked APK: 200 hits, all reviewed as license URLs, XML namespaces and strings inside libflutter, libapp and classes.dex (words such as "password" and "staging"). No keys, tokens, private-key blocks or .env content found. Values were redacted by the script.
- 16 KB page alignment: `zipalign -c -P 16` reports OK for every native library checked in the arm64 APK.
- Play AAB manifest: no REQUEST_INSTALL_PACKAGES, not debuggable.
- Runtime on emulator: 0 fatal exceptions or ANRs in 4,881 logcat lines; process kill then relaunch recovers; airplane-mode launch works; Library at 2.0 font scale renders without overflow.

Appendix B - Inventories

B1. Permissions (merged manifest, sideload arm64)

Permission	Source	Needed for	Runtime prompt	Verdict
INTERNET	app	book lookup, covers, update check	no	justified
CAMERA	app	barcode scanner	yes, on first scan	justified
USE_BIOMETRIC	app	fingerprint unlock	no	justified
USE_FINGERPRINT	local_auth (legacy)	same, older Android	no	justified, missing from policy
POST_NOTIFICATIONS	app	reminders and backup-problem notice	yes (Android 13+)	justified
REQUEST_INSTALL_PACKAGES	sideload flavor only	install a downloaded update	special access	justified, absent from Play build
ACCESS_NETWORK_STATE	plugin	backup and lookup network checks	no	justified
WAKE_LOCK	WorkManager	scheduled work	no	missing from policy
RECEIVE_BOOT_COMPLETED	WorkManager / notifications	re-schedule reminders after reboot	no	missing from policy
FOREGROUND_SERVICE	WorkManager	long background work	no	confirm needed; missing from policy
VIBRATE	notifications	notification vibration	no	missing from policy
com.axialforge.pagemark.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	AndroidX	internal receiver protection	no	normal

Removed on purpose with `tools:node="remove"`: READ_EXTERNAL_STORAGE, READ_MEDIA_IMAGES, READ_MEDIA_VIDEO, READ_MEDIA_AUDIO. Not present: location, contacts, microphone, SMS, storage, accounts, advertising ID. On the Android 17 emulator the system also showed ACCESS_LOCAL_NETWORK as granted automatically for this older target SDK; the app does not declare it.

B2. Application components

Component	Exported	Guard	Notes
MainActivity	yes	none needed	MAIN/LAUNCHER and a SEND text/plain share target; shared text is capped, whitelisted and only pre-fills the search box
PagemarkWidgetProvider (receiver)	no		home-screen widget, main library only
androidx.glance GlanceRemoteViewsService	yes	BIND_REMOTEVIEWS	system-only binding
WorkManager SystemJobService	yes	BIND_JOB_SERVICE	system-only binding
WorkManager DiagnosticsReceiver, profileinstaller ProfileInstallReceiver	yes	DUMP	system and shell only
ML Kit, content providers and other library components	no		

Application flags: allowBackup=false, fullBackupContent=false, dataExtractionRules exclude root, file, database, sharedpref and external for both cloud backup and device transfer, not debuggable, extractNativeLibs=false.

B3. Network endpoints (all HTTPS, from source)

Host	Purpose	Trigger	Data sent
openlibrary.org, covers.openlibrary.org	search, ISBN lookup, covers	user search or scan, cover display	query text or ISBN, IP address
www.googleapis.com (Google Books)	second lookup source, can be switched off	user search	query text
api.github.com, github.com	update check and APK download (sideload flavor)	at most daily, switchable	public release request only

No cleartext configuration, no custom TrustManager or HostnameVerifier, no certificate pinning (not needed for public book data). A test in the repository enforces the host allow-list (`promiseHosts`).

B4. Data-flow inventory

Data	Where stored	Encrypted at rest	Leaves device?	Recipient	Deletion
Books, reads, ratings, thoughts, quotes, shelves	SQLite in app-private storage	no (owner-accepted, MOB-SEC-0006)	only through exports and backups the user saves	user-chosen location	uninstall, or in-app delete and Trash
Locked library	same database, `locked` flag	no	in the full-database backup only, optionally AES-256-GCM encrypted	user-chosen location	in-app
PIN hash, salt, recovery hash	flutter_secure_storage (Keystore backed)	yes	no		clear lock, or uninstall
Backup passphrase	flutter_secure_storage	yes	no		uninstall
Settings, filters, timer state	SharedPreferences	no (non-sensitive)	no		uninstall
Event log (200 KB cap, SQL details scrubbed)	app files	no	only if the user shares a problem report	user	in-app
Cover images	cache and `local:` cover store; locked covers in a separate cache wiped on lock	no	no		in-app
Search text and ISBN	memory	n/a	yes, to the lookup hosts above	Open Library, Google Books	not retained by the app
Camera frames	memory only	n/a	not stored or sent by the app; the ML Kit library may report technical data (MOB-PRV-0002)		

Accounts, advertising IDs, analytics, crash reporting, location and contacts: none. Account deletion requirements: not applicable (no accounts).

B5. Cryptography and local security

Item	Finding
PIN	PBKDF2-HMAC-SHA256, 150,000 iterations, random 16-byte salt per install, constant-time compare, attempt limiting with lockout up to 1 hour (clock-skew capped), recovery code stored as a hash; legacy SHA-256 PIN upgraded on use

Item	Finding
Backup encryption	"PGMKENC1": AES-256-GCM, 16-byte random salt, 12-byte nonce, key from PBKDF2-SHA256 at 150,000 iterations
Secrets in source	none found; signing config reads an untracked `key.properties`
Screen capture	FLAG_SECURE while the locked library is open; the vault relocks on pause (static review; recents thumbnail not verified at runtime)
Locked-library leaks	exports, stats, widget and suggestions filter on `locked`; covered by repository tests

B6. Dependencies (pub.dev, from pubspec.lock)

187 packages. OSV result: no known vulnerabilities.

Direct packages of note: drift 2.35.2, sqlite3 3.7.0, sqlite3_flutter_libs 0.6.0+eol (end-of-life, MOB-AND-0004), http 1.6.0, local_auth 3.0.2, flutter_secure_storage 11.2.0, cryptography 2.9.0, mobile_scanner 7.4.2 (Google ML Kit), home_widget 0.10.0, flutter_local_notifications 22.3.2, file_picker 13.1.0, share_plus 13.3.1, open_filex 4.7.0, image_picker 1.2.4, pdf 3.13.1, archive 4.5.1, cached_network_image 4.0.4, package_info_plus 10.2.2, device_info_plus 13.3.0, android_intent_plus 6.1.0. No analytics, advertising, crash-reporting or payment SDK.

Native libraries in the arm64 APK: libflutter, libapp, libsqlite3, libbarhopper_v3 (ML Kit barcode), libdartjni, libimage_processing_util_jni, libsurface_util_jni, libdatastore_shared_counter.

Full package list (SBOM)

Package	Version
_fe_analyzer_shared	109.0.0
analyzer	14.5.0
android_file_picker	2.0.1
android_intent_plus	6.1.0
archive	4.5.1
args	2.7.0
async	2.13.1
barcode	2.2.9
bidi	2.0.13
boolean_selector	2.1.2
build	4.0.11
build_config	1.3.3
build_daemon	4.1.6
build_runner	2.16.2
built_collection	5.1.2
built_value	8.13.0
cached_network_image	4.0.4
cached_network_image_platform_interface	5.0.3
cached_network_image_web	2.0.3
characters	1.4.1
charcode	1.4.0
checked_yaml	2.0.4
cli_util	0.6.0

Package	Version
clock	1.1.3
code_assets	2.1.0
collection	1.19.1
convert	3.1.2
cross_file	0.3.5+5
crypto	3.0.7
cryptography	2.9.0
cryptography_flutter	2.3.4
csv	8.0.0
cupertino_icons	1.0.9
cupertino_ui	1.1.2
dart_style	3.1.13
dbus	0.8.0
device_info_plus	13.3.0
device_info_plus_platform_interface	8.1.0
drift	2.35.2
drift_dev	2.35.1
fake_async	1.3.3
ffi	2.2.0
ffi_leak_tracker	0.1.2
file	7.0.1
file_picker	13.1.0
file_picker_darwin	2.2.0
file_picker_linux	2.0.2
file_picker_platform_interface	4.1.0
file_picker_web	4.1.0
file_selector_linux	0.9.4+1
file_selector_macos	0.9.5+1
file_selector_platform_interface	2.7.0
file_selector_windows	0.9.3+6
fixnum	1.1.1
flutter	0.0.0
flutter_cache_manager	3.4.5
flutter_lints	6.0.0
flutter_local_notifications	22.3.2
flutter_local_notifications_linux	8.0.2
flutter_local_notifications_platform_interface	12.2.0
flutter_local_notifications_web	1.0.0
flutter_local_notifications_windows	3.1.1
flutter_localizations	0.0.0
flutter_plugin_android_lifecycle	2.0.35
flutter_secure_storage	11.2.0
flutter_secure_storage_darwin	0.4.3
flutter_secure_storage_linux	3.0.3

Package	Version
flutter_secure_storage_platform_interface	2.1.1
flutter_secure_storage_web	2.1.1
flutter_secure_storage_windows	4.2.2
flutter_test	0.0.0
flutter_web_plugins	0.0.0
glob	2.2.0
graphs	2.3.2
home_widget	0.10.0
hooks	2.2.0
http	1.6.0
http_multi_server	3.2.2
http_parser	4.1.2
image	4.12.0
image_picker	1.2.4
image_picker_android	0.8.13+25
image_picker_for_web	3.1.1
image_picker_ios	0.8.13+10
image_picker_linux	0.2.2
image_picker_macos	0.2.2+1
image_picker_platform_interface	2.11.2
image_picker_windows	0.2.2
intl	0.20.3
io	1.1.0
jni	1.1.0
jni_flutter	1.0.4+1
jni_util	1.0.0
json_annotation	4.12.0
leak_tracker	11.0.2
leak_tracker_flutter_testing	3.0.10
leak_tracker_testing	3.0.2
lints	6.1.0
local_auth	3.0.2
local_auth_android	2.2.0
local_auth_darwin	2.0.4
local_auth_platform_interface	1.1.0
local_auth_windows	2.0.2
logging	1.3.0
matcher	0.12.20
material_color_utilities	0.13.0
material_ui	1.6.0
meta	1.19.0
mime	2.1.0
mobile_scanner	7.4.2
native_toolchain_c	0.19.5

Package	Version
objective_c	9.6.2
octo_image	2.1.2
open_filex	4.7.0
package_config	3.0.0
package_info_plus	10.2.2
package_info_plus_platform_interface	4.1.0
path	1.9.1
path_parsing	1.1.0
path_provider	2.1.6
path_provider_android	2.3.1
path_provider_foundation	2.6.0
path_provider_linux	2.2.2
path_provider_platform_interface	2.1.3
path_provider_windows	2.3.0
pdf	3.13.1
petitparser	7.1.0
platform	3.2.0
plugin_platform_interface	2.1.8
pool	1.5.3
posix	6.5.2
process	5.0.6
pub_semver	2.2.1
pubspec_parse	1.6.0
qr	3.0.2
recase	4.1.0
record_use	1.1.1
rxdart	0.28.0
share_plus	13.3.1
share_plus_platform_interface	7.2.0
shared_preferences	2.5.6
shared_preferences_android	2.4.28
shared_preferences_foundation	2.5.7
shared_preferences_linux	2.4.1
shared_preferences_platform_interface	2.4.2
shared_preferences_web	2.4.3
shared_preferences_windows	2.4.1
shelf	1.4.2
shelf_web_socket	3.0.0
sky_engine	0.0.0
source_gen	4.3.0
source_span	1.10.2
sqflite	2.4.4+1
sqflite_android	2.4.4
sqflite_common	2.5.13+1

Package	Version
sqflite_darwin	2.4.4
sqflite_platform_interface	2.4.2
sqlite3	3.7.0
sqlite3_flutter_libs	0.6.0+eol
sqlparser	0.45.0
stack_trace	1.12.2
stream_channel	2.1.4
stream_transform	2.1.2
string_scanner	1.4.1
synchronized	3.4.2
term_glyph	1.2.2
test_api	0.7.12
timezone	0.11.2
typed_data	1.4.0
url_launcher_linux	3.2.3
url_launcher_platform_interface	2.3.2
url_launcher_web	2.4.3
url_launcher_windows	3.1.6
uuid	4.6.0
vector_math	2.4.3
vm_service	15.3.0
watcher	1.2.1
web	1.1.1
web_socket	1.0.1
web_socket_channel	3.0.3
win32	6.4.0
win32_registry	3.0.3
windows_file_picker	2.0.1
xdg_directories	1.1.0
xml	7.1.0
yaml	3.1.4
yaml_edit	2.2.4

Appendix C - Store disclosure, device matrix and open items

C1. Store disclosure diff (Google Play Data safety)

Data type	Declared in docs/PLAY_STORE.md	Observed	Match
Search queries, app activity	sent to Open Library and Google Books, optional, not linked to identity	confirmed in source (lookup.dart)	yes
Personal info, financial, health, location, contacts, messages, photos, audio	none	none found in the manifest or source	yes
Camera images	processed on device, not stored or sent	the app never stores or sends them; the ML Kit library may report technical info to Google per the privacy policy	UNVERIFIED (MOB-PRV-0002)
Device or other IDs, diagnostics	none	not verified at runtime for the ML Kit component	UNVERIFIED
Encryption in transit	HTTPS	all hosts HTTPS, no cleartext	yes
Deletion	clear storage, delete backups	no accounts; uninstall removes app data	yes
Permissions in the privacy policy	internet, camera, biometric, install packages, notifications, background work and network state	five more in the merged manifest	NO (MOB-PRV-0001)
Listing claims "no account, no ads, no tracking, no AI, never collects your data"		no accounts, ads, analytics or AI SDKs found; lookups expose search text and IP to third parties, which the policy discloses	yes, with the ML Kit caveat

C2. Play checklist status

Item	State
Target API meets the requirement (36)	PASS
64-bit and AAB	PASS (arm64 and armv7 APKs, AAB built)
16 KB page alignment	PASS
Installer permission absent from the Play flavor	PASS
Data safety form	MANUAL REVIEW
Privacy policy URL (public GitHub page)	present; content needs the MOB-PRV-0001 fix
Content rating and target audience (13+)	MANUAL REVIEW (answers drafted)
Account deletion	N/A, no accounts
Ads declaration	none
Restricted-permission declarations	none needed (no SMS, call log, background location, all-files access)
Feature graphic 1024x500 and current screenshots	OPEN (MOB-STORE-0005)
Pre-launch report	NOT RUN (needs Play Console); the test robot cannot pass a PIN lock

C3. Device and OS matrix actually exercised

Cell	Exercised
Pixel_10_Pro_XL_test emulator, Android 17, x86_64 with arm64 translation, 1344x2992	yes: install-over, launch, offline, 2.0 font scale, process kill
Real phone (any)	NO

Cell	Exercised
Low-end device, Samsung, tablet or foldable	NO (a tablet two-pane layout was exercised in the earlier 1.0.0 gate audit, emulator only)
Min SDK 24 device	NO
Android 13 notification prompt, biometric hardware, camera hardware	NO

C4. Manual review queue

1. Capture the barcode scanner's network traffic once on a device, then fix or confirm the Data safety form (MOB-PRV-0002).
2. Update PRIVACY.md (permission list and Drive wording) and decide whether FOREGROUND_SERVICE can be stripped (MOB-PRV-0001).
3. Real-phone pass from docs/release-audit/MANUAL_TEST_PLAN.md: reading timer, folder backup to a real cloud folder, fingerprint, notifications on Android 13+, camera scan, share sheet, widget, TalkBack, in-app updater install, upgrade from the published 0.13.0 APK.
4. Check the recents-screen thumbnail and clipboard behavior with the locked library open.
5. Fresh install, uninstall and reinstall, and what survives (keystore data, backup folder).
6. Confirm on a device that the 0.13.1 fixes for PGM-300 (Stats width), PGM-302 (genre noise) and PGM-320 (stuck timer) work; they are marked ready for verification in the repository audit.
7. Real-phone performance: cold start and scrolling a large library (the earlier 5,000-book test was emulator only).
8. Create the Play listing assets and run Play's pre-launch report.

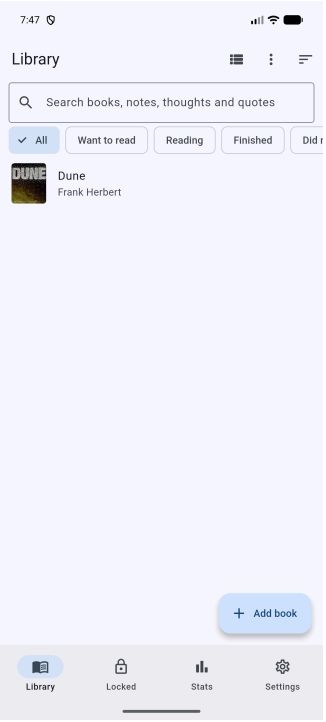
C5. What is missing for a clean GO

Items 1, 2 and 3 above. With those closed and no new S2 or higher findings, the Play decision moves to GO; the sideload APKs are already a conditional go.

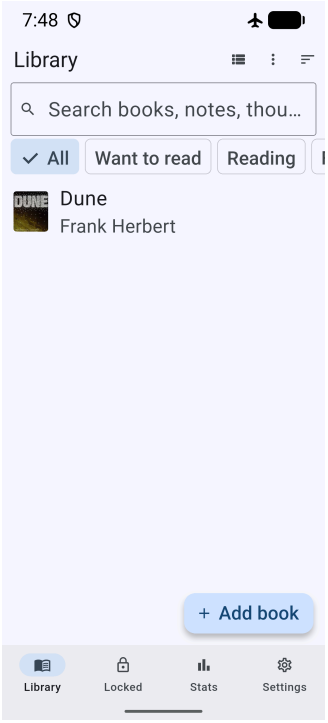
Appendix D - Runtime evidence

Emulator: Pixel_10_Pro_XL_test, Android 17, x86_64. Pagemark 0.13.1 (2022) was installed over an existing install with the same signing key; the library kept its data.

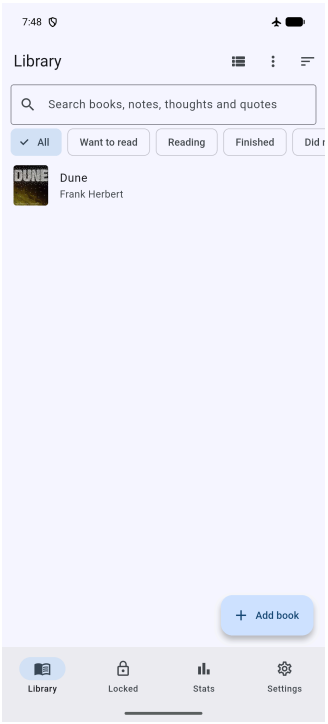
Measure	Result
Cold start (am start -W TotalTime), 3 runs	4,409 ms, 3,021 ms, 2,795 ms (ABI translation inflates this)
Offline launch (airplane mode)	3,086 ms, app opened normally
Memory after launch	TOTAL PSS 230,927 KB, RSS 376,772 KB, Java heap 5.6 MB, native heap 27.5 MB
First-frame render	1 frame, 23 ms (not a useful jank measure)
Fatal exceptions or ANRs in logcat	0 of 4,881 lines; only SELinux max_map_count notices and a CPU-variant warning from the emulator
Debugger access to private data	`run-as` refused: package not debuggable
Kill and relaunch	process restarted, no crash



Library on launch (default text size)



Library at 2.0 font scale, airplane mode on



Library launched in airplane mode

Test suite tail (flutter test)

`00:16 +295 ~1: All tests passed!` (295 passed, 1 skipped). `flutter analyze`: No issues found.